WEEK 6 OF 15 · STAGE 1: FOUNDATIONS

The scenario framework

SSPs, RCPs and how to read a scenario chart

New R skill tidyr: long and wide data; joins; reports in R Markdown

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 — Foundations

- Week 1: Why model the future?
- Week 2: Climate primer
- Week 3: Energy primer
- Week 4: Economics primer
- Week 5: Cost-benefit IAMs: DICE
- **Week 6: The scenario framework**

Stage 2 — GCAM in depth

- Week 7: Inside GCAM
- Week 8: Energy in GCAM
- Week 9: Land, water and bioenergy
- Week 10: Emissions and policy

Stage 3 — Advanced methods

- Week 11: Optimisation IAMs
- Week 12: Why models disagree
- Week 13: Uncertainty
- Week 14: Frontiers and limits
- Week 15: Projects and review

TODAY'S GOALS

By the end of today you can

1. Explain what a scenario is and why researchers share one set
2. Decode a label such as SSP2-4.5 into a society and a forcing level
3. Ask four questions of any scenario chart
4. Reshape wide data to long, join tables and knit an R Markdown report

? **Big question** The highest emissions scenario was long called 'business as usual'. Was it?

Was it 'business as usual'?

The highest emissions scenario, SSP5-8.5, was long called 'business as usual'. It describes a world that burns far more coal than current trends suggest.



What does a scenario's name tell you, and what does it not?

A scenario is a choose-your-own-adventure book



Each story describes how society might develop. A spreadsheet is stapled to every page: population, income, technology costs. Models then turn the story into energy use, emissions and warming.

What it means for models

1. A scenario is a consistent what-if story with numbers
2. It is not a forecast
3. Shared stories let different models be compared

KEY IDEA

Five SSP stories about society

1

SSP1 Sustainability

Green growth, education, low inequality. Easy to cut emissions and to adapt.

2

SSP2 Middle of the road

Trends roughly continue. Moderate challenges on both sides.

3

SSP3 Regional rivalry

Countries turn inward, slow growth, high population. Hard to cut and to adapt.

4

SSP4 and SSP5

SSP4: deep inequality. SSP5: fast fossil-fuelled growth, hard to cut but easier to adapt.

COMPARE

Two labels, two jobs

SSP: society

- Population, income, technology, inequality
- Values: SSP1 to SSP5
- Fixes the starting story

RCP: climate outcome

- Radiative forcing in 2100, in W/m^2
- Values: 1.9, 2.6, 4.5, 7.0, 8.5
- Higher number means more warming

SIDE BY SIDE

The scenario matrix: society × climate

Society	1.9	2.6	4.5	7.0 / 8.5
SSP1 Sustainability	SSP1-1.9	SSP1-2.6		
SSP2 Middle of the road			SSP2-4.5	
SSP3 Regional rivalry				SSP3-7.0
SSP5 Fossil-fuelled				SSP5-8.5

Columns: forcing in 2100 (W/m^2). Filled cells: the five headline scenarios in IPCC AR6 climate runs.

Quick check

What does the 7.0 in SSP3-7.0 mean?

A 7.0 °C of warming

B 7.0 W/m² of forcing in 2100

C 7 billion people

D 7.0 Gt of CO2 a year

Quick check: answer

What does the 7.0 in SSP3-7.0 mean?

A. 7.0 °C of warming

B. 7.0 W/m² of forcing in 2100 ✓

C. 7 billion people

D. 7.0 Gt of CO2 a year

SIDE BY SIDE

IPCC AR6 sorts pathways by warming

Category	Warming outcome
C1	1.5 °C (>50%), little or no overshoot
C2	Back to 1.5 °C after high overshoot
C3	Below 2 °C (>67% chance)
C4 to C6	2 °C, 2.5 °C, 3 °C (>50%)
C7 and C8	Below 4 °C, or above 4 °C

About 1,200 pathways in the 2022 report. Check category wording against AR6 WGIII before teaching.

HOW IT WORKS

Four questions for any scenario chart

1

Which model?

The same policy gives different results in different models

2

Which society?

Population, income and costs drive the result

3

Reference or policy?

A reference case is not a prediction

4

What is hidden?

A spread across models is not a probability

Put a real AR6 chart on screen and answer these in pairs.

Worked example: wide to long by hand

Problem Two models, A and B, each report CO2 in 2030 and 2050 in wide format: A: 30, 5. B: 26, 2. Make it long, and find A's cut.

1. Wide: 2 rows × 2 year columns = 4 numbers
2. Long: one row per number, so 4 rows with columns year and value
3. Rows: A 2030 30; A 2050 5; B 2030 26; B 2050 2
4. A's cut: (30 − 5) ÷ 30 = 25 ÷ 30 ≈ 0.83, so 83% lower in 2050

Answer

4 long rows; model A cuts CO2 by about 83% from 2030 to 2050

WATCH OUT

Common mix-ups

Not quite	Better
SSP2-4.5 means 4.5 °C of warming.	4.5 is forcing in W/m². AR6 assesses SSP2-4.5 at roughly 2.7 °C by 2100.
30 of 40 models show it, so there is a 75% chance.	Models are not a random sample of futures. Counting them does not give a probability.
The reference scenario is what will happen.	It is what follows with no new policy, under its assumptions. It is a yardstick, not a forecast.

Reshape and join in R

- pivot_longer turns year columns into rows
- matches("^[0-9]{4}$") means 'all the year columns'
- left_join adds each pathway's category by matching keys

```
long <- ar6 |>
  filter(Variable == "Emissions|CO2") |>
  pivot_longer(matches("^[0-9]{4}$"),
               names_to = "year",
               values_to = "mt")

both <- long |>
  left_join(meta,
            by = c("Model", "Scenario"))
```

NEW WORDS THIS WEEK

Words to add to the class glossary

SSP Shared Socioeconomic Pathway: a story about how society develops (population, income, technology). Five stories, SSP1 to SSP5.

RCP Representative Concentration Pathway: a climate outcome, named by its radiative forcing in 2100 in W/m^2, such as 4.5.

Scenario matrix A grid with SSPs as rows and forcing levels as columns. Each filled cell is one combined scenario.

AR6 category The IPCC's sorting of pathways by warming outcome, from C1 (1.5 °C) to C8 (above 4 °C).

Long data A table with one row per observation, such as one row per model, scenario and year. ggplot needs it.

HANDS-ON LAB

Lab 6: reading AR6 scenarios

Reshape AR6 scenario data from wide to long, attach each pathway's warming category, plot CO2 by category, and knit an R Markdown report.

Script: labs/lab6.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Read and reshape

Keep only total CO2, turn year columns into rows, and convert Mt to Gt.

```
ar6 <- read_csv("data/ar6_world.csv")
long <- ar6 |>
  filter(Variable == "Emissions|CO2") |>
  pivot_longer(
    matches("^[0-9]{4}$"),
    names_to = "year",
    values_to = "mt"
  ) |>
  mutate(year = as.integer(year),
         gt = mt / 1000)
```

Join the categories

meta has one row per pathway with its category. left_join matches on Model and Scenario.

```
meta <- read_csv("data/ar6_meta.csv")
both <- long |>
  left_join(meta,
    by = c("Model", "Scenario")) |>
  filter(Category %in% c("C1", "C3", "C7")) |>
  mutate(run = paste(Model, Scenario))
```

Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Plot one panel per category

group = run stops ggplot from joining all points into one zigzag. alpha makes lines see-through.

```
ggplot(both, aes(year, gt, group = run)) +
  geom_line(alpha = 0.2) +
  facet_wrap(~ Category)
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Knit a report



File > New File > R Markdown. Paste the code into chunks, write one sentence under each chart, and press Knit.

Did it work?

You should see

- long has one row per model, scenario and year
- Your chart has three panels: C1, C3 and C7
- The report knits to HTML without errors

Worksheet questions

1. Describe the C1 panel in two sentences: when do emissions reach about zero?
2. Apply the four questions to your chart. Which is hardest to answer from the data alone?
3. Why can't the share of C1 pathways be read as the chance of reaching 1.5 °C?

RECAP

Take-home messages

1. A scenario is a consistent what-if story, not a forecast
2. SSPs describe society; RCP numbers are forcing in W/m^2 in 2100, not °C
3. AR6 sorts about 1,200 pathways into categories C1 to C8 by warming
4. Ask four questions of any scenario chart; a range across models is not a probability

Next: week 7

Week 7 opens a full process-based IAM: GCAM, its regions and markets, and how to pull its results into R.

Practise

Interactive course, week 6: try the matrix explorer and the 5-question quiz.

Readings: Carbon Brief (2018), 'Explainer: How Shared Socioeconomic Pathways explore future climate change'; Optional: Riahi et al. (2017), Global Environmental Change; R for Data Science: data tidying and joins chapters